Machine learning for porous materials

ML MolSim foundations

Advanced Methods in Digital Chemistry
11 November 2026
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Splitting data, scaling, cross-validation, kernels

Recorded summary (Loom)

Our first marimo notebook

00-getting-started.py

Change one input. Predict which cells will change.

```
concentration = 2.0
prediction = slope * concentration + intercept
```

Save your own notebook copy.

Where machine learning enters

Structure — Interactions — Simulation — **Uptake**

Descriptors — Learned surrogate — **Predicted uptake**

Our labels come from completed simulations.

What are we trying to predict?

X —— **Model** —— **y**

Geometric descriptors

Simulated CO_2 uptake

298 K, 16 bar, mmol/g

A structure becomes a row of numbers

Structure

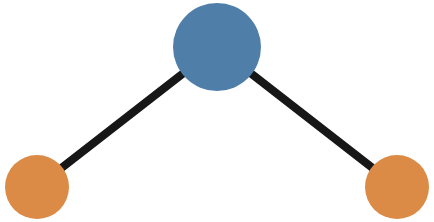
Descriptors

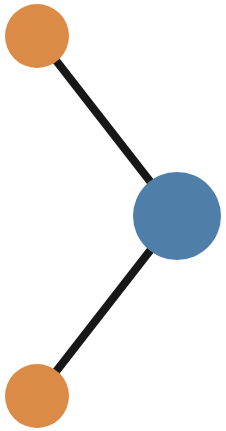
Material	Density	Pore volume	Pore size
A			
B			

Which chemical information have we discarded?

Prior knowledge constrains the model

The same molecule, rotated





Coordinates change.

Energy stays the same.

Which transformations should our representation ignore?

A split defines the prediction task

Training: 900	Validation: 300	Test: 300
Fit	Choose	Assess

What counts as a genuinely new material?

Fitting adjusts the model parameters

$$\hat{y} = w_0 + w_1 x$$

Predictions



Training loss



Compare with training targets, then update the parameters.

A baseline and one error calculation

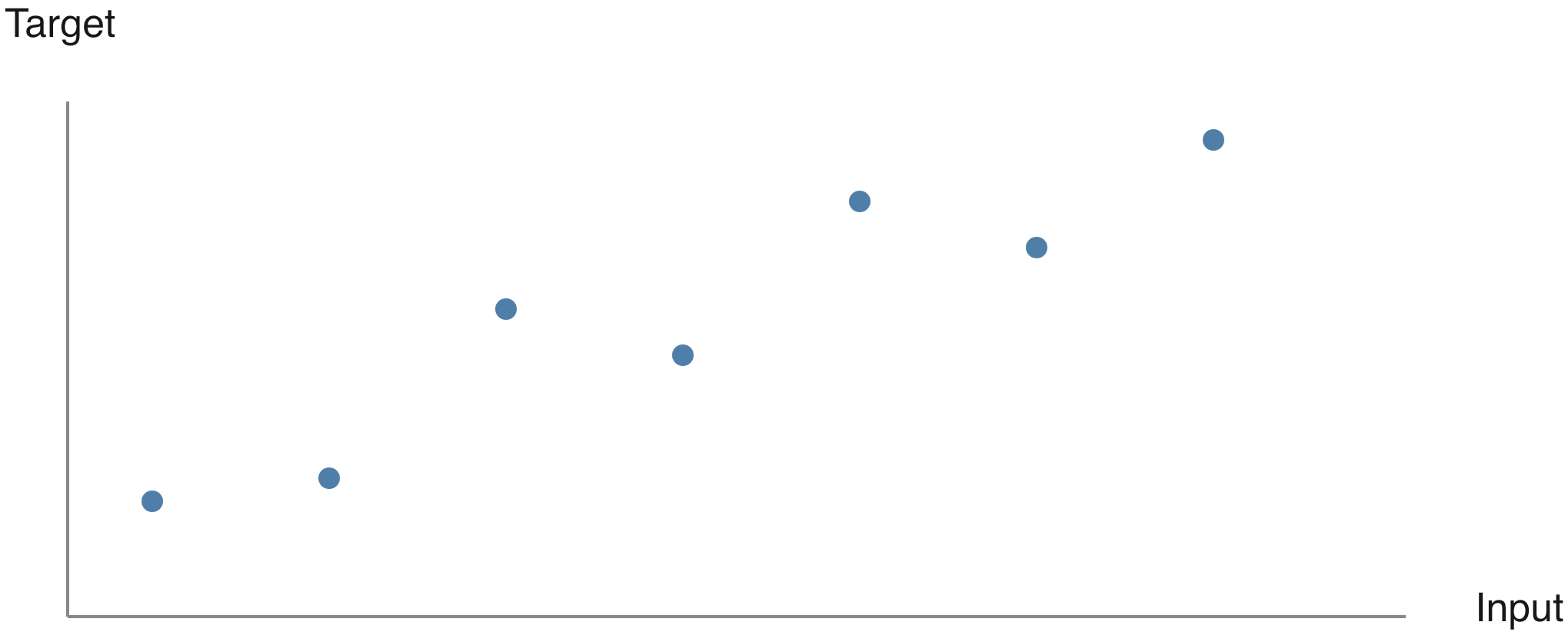
Baseline: median training uptake

MAE = mean absolute prediction error

Residuals: –2, 0, +1 mmol/g

MAE =

Training error can keep falling



Which curve will predict the next observation?

Regularisation constrains the fit

Fitting objective = training loss + λ × penalty

Small λ

Large λ

Validation helps choose the strength λ.

Validation chooses the settings

Training data

Fit model parameters

Validation data

Compare candidate settings

Test data stay outside this decision.

Cross-validation repeats the comparison

Fit 1	Validate	Train	Train	Train	Train
Fit 2	Train	Validate	Train	Train	Train
Fit 3	Train	Train	Validate	Train	Train
Fit 4	Train	Train	Train	Validate	Train
Fit 5	Train	Train	Train	Train	Validate

Held-out scores

Mean =

The separate test set remains untouched.

Nonlinear features can bend a linear model

Features: x, x^2

$$\hat{y} = w_0 + w_1 x + w_2 x^2$$

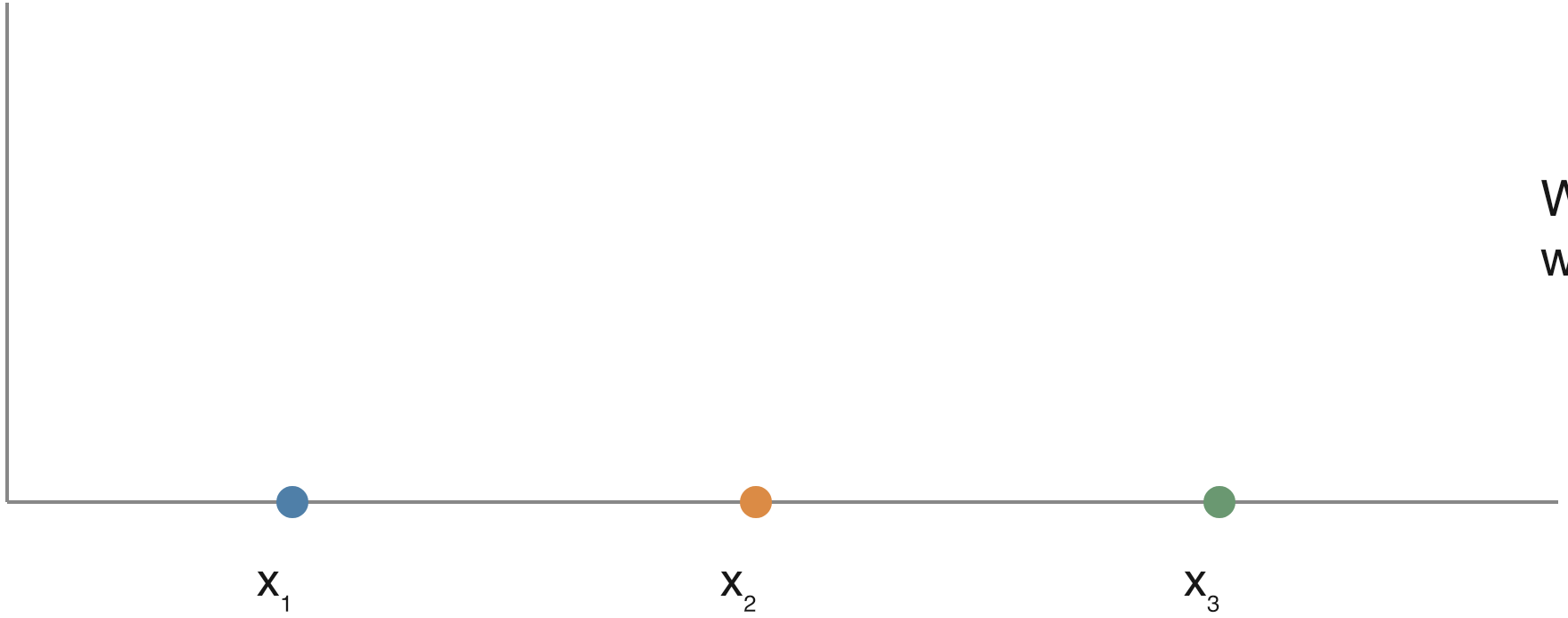


Linear in the
fitted coefficients

A kernel gives us a way to work with a richer feature space.

A kernel describes similarity

One contribution centred on each training input



Width changes
what “near” means.

Units change numerical distances

Difference in density	**0.1 g/cm^3**
Difference in surface area	**100 m^2/g**

Squared distance =

Standardisation uses training statistics

$$z = (x - \mu) / \sigma$$

One mean and standard deviation for each feature

μ and σ come from the training data.

Validation and test use the same transformation.

Preprocessing belongs inside each fold

Training fold —— Fit scaler —— Fit model

Validation fold —— Transform —— Predict and score

The final test answers a narrower question

Freeze the workflow. Then assess the test predictions.

Which materials does this result describe?

Error metric and units

Hands-on with the ML MolSim material

03-mof-uptake.py

Baseline

Unscaled and scaled KRR

Record the validation comparison before the test.

Supported practical: 2 December, 12:00–16:00

A project starts with a chemical question

What would you like to find out?

Which data could you use?

What would be a useful first comparison?

Add an idea to the document in the course email.